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'How can we check the evidence presented by GenAI's response?': students' epistemic vigilance of their co-constructed scientific explanations with GenAI

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ABSTRACT

This study describes how middle school science students enacted epistemic vigilance while co-constructing scientific explanations with generative artificial intelligence (GenAI). Using a qualitative research design, the study investigated the interactions of eight middle school students with Canva and ChatGPT during two science lessons on plant growth and the classification of fungi. To support their epistemic vigilance, two explicit epistemic scaffolds were implemented: the criteria for good scientific explanations as epistemic criteria and epistemic question prompts as reflective guides. Data including transcribed and translated video recordings of student-GenAI interactions, student-generated digital artifacts, and semi-structured interviews were analyzed using a multi-layered analytical framework. Findings indicate that students enacted epistemic vigilance through three complementary evaluative practices: evaluating the scientific quality of GenAI-generated explanations, verifying the credibility of information sources, and reflecting on their own understanding and assumptions while interpreting AI outputs. The epistemic criteria primarily scaffolded students' content evaluation while indirectly supporting source and receiver evaluation. Conversely, the epistemic question prompts explicitly scaffolded source and receiver evaluation while remaining grounded in evaluating their co-constructed scientific explanation quality. These findings suggest that within their cognitive levels, middle school students may engage in epistemic vigilance with explicit epistemic scaffolding. Limitations are further discussed.

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KEYWORDS

Explicit epistemic scaffolding; epistemic vigilance; scientific explanations; student-GenAI collaboration

Introduction

The Next Generation Science Standards (NGSS Lead States, 2013) emphasise students' participation in epistemic practices of science, including constructing explanations, evaluating evidence, and engaging in scientific reasoning. Concurrently, generative artificial

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intelligence (GenAI) has increasingly influenced science and science education (AAAS, 2021; Alasadi & Baiz, 2023; Cooper, 2023). Erduran and Levrini (2024) argued that meaningful engagement with GenAI is necessary to prevent students' alienation and support their informed participation in AI-mediated knowledge construction. However, effective participation requires students not only to generate scientific explanations with GenAI but also to critically evaluate the knowledge it produces.

Recent studies have shown that GenAI can support students' reflection, explanation revision, scientific reasoning, and epistemic understanding (Chang et al., 2026; Sirnoorkar et al., 2024; Tang et al., 2024; Tang & Putra, 2026). Nevertheless, these studies have primarily examined how GenAI enhances learning outcomes, with limited attention to how students evaluate AI-generated explanations during the co-construction process. This issue is important because GenAI can produce scientifically plausible yet inaccurate or incomplete explanations (Anderson et al., 2024; Doshi & Hauser, 2024; Mishra et al., 2023).

Addressing this challenge requires epistemic vigilance, defined as learners' capacity to critically evaluate communicated information before accepting it (Sperber et al., 2010). Although epistemic vigilance has been conceptualised in science education (Bielik & Krell, 2025; Tseng et al., 2021), little empirical research has examined how students enact it while co-constructing scientific explanations with GenAI, particularly when supported by explicit epistemic scaffolds. This study addresses this gap by investigating how middle school students enact epistemic vigilance during GenAI-supported co-construction of scientific explanations using epistemic criteria and epistemic question prompts.

Literature review

Because GenAI has the potential to enhance learning, creativity, and knowledge production, interest in using it to teach science has grown significantly (Mishra et al., 2023). According to O'Toole and Horv'y's (2024) study, conversational GenAI chatbots can aid metacognitive processes by making information retrieval and problem solving easier. Additionally, GenAI has potential for mechanical reasoning in physics discourse (Sirnoorkar et al., 2024), inquiry-based learning (Erduran & Levrini, 2024; Penn & Ramnarain, 2025), and evaluation of epistemic understanding (Chang et al., 2026). By offering immediate feedback and tailored guidance via student prompts, GenAI chatbots promote self-regulated learning (Ng et al., 2024).

Despite these potentials, researchers raised the issue of hallucination, which brings an emphasis on how humans, including students, must remain central agents in AI-mediated learning environments. Students' agency involves their participation, ownership of outcomes, and mental effort while engaged with GenAI (Draxler et al., 2024; Guo et al., 2025). This framing revolves around the idea that although GenAI can accomplish tasks efficiently, they still need humans to demonstrate exploratory thinking patterns (Boussieux et al., 2024). Accordingly, GenAI cannot replace human judgment because the responsibility for validating knowledge remains with humans (Hemmer et al., 2021; Oh & Lee, 2025). Thus, in this time when AI has shaped science learning, students should be exposed to human-AI collaboration while being trained to be vigilant when to trust, question, or build upon AI-generated outputs (Hofßbach & Isaksen, 2025).

However, with increasingly studies focused on human-AI collaboration, creativity, reasoning, adaptive learning, or knowledge co-production (Cheah et al., 2025; Lubbe et al., 2025), no studies have explored epistemic vigilance within science learning contexts. More specifically, little attention has been given to how students enact epistemic vigilance when co-constructing scientific explanations with GenAI. Thus, epistemic vigilance during students' construction of scientific explanation with GenAI remains under-explored today.

Theoretical framework

Epistemic vigilance

Epistemic vigilance, as earlier defined, is composed of three aspects: content evaluation, source evaluation, and receiver evaluation (Bielik & Krell, 2025; Pimentel, 2025; Sperber et al., 2010), each with some principles to be considered during evaluation (Figure 1). Content evaluation may be assessed through three complementary principles: plausibility, one-sidedness, and context (Duncan et al., 2018; Tseng et al., 2021). In relation to learning, it requires students to determine whether explanations are evidence-based, credible, or logically coherent (Allchin, 2023; Osborne & Pimentel, 2023). It also involves identifying one-sided arguments, missing information, and alternative explanations that may challenge scientific claims (Barzilai & Chinn, 2018). Within science learning, this aspect is connected to constructing scientific explanations because students are expected

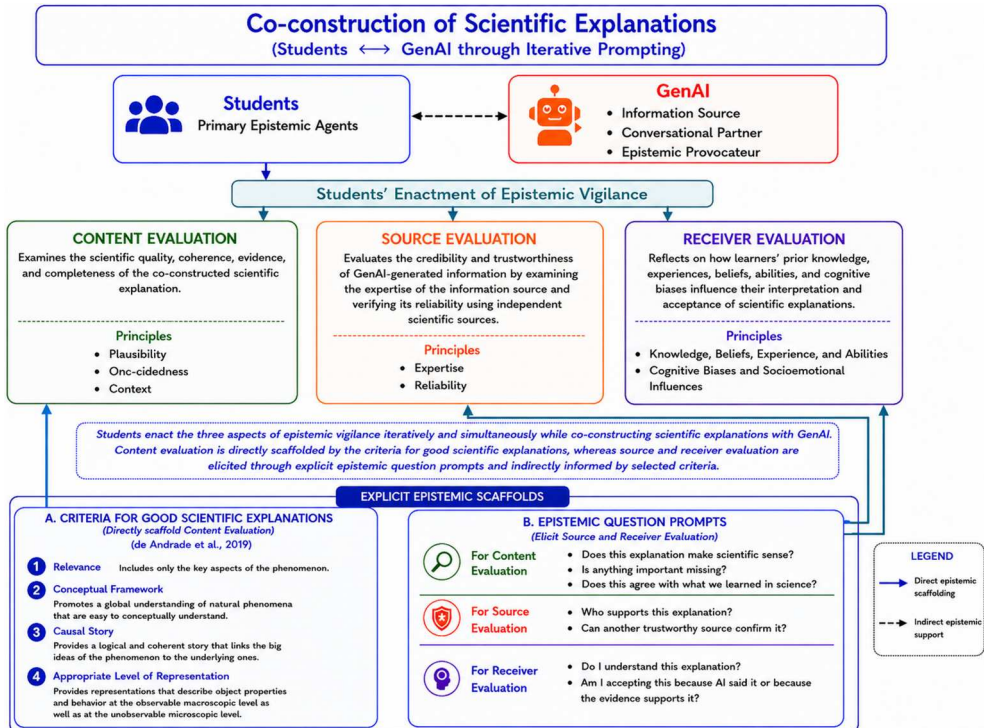


Figure 1. Conceptual framework of the study. Note: This diagram design was enhanced by GenAI.

to evaluate evidence, scrutinise reasoning, and distinguish scientific explanations from unsupported rhetoric.

The second aspect is source evaluation, which focuses on assessing the trustworthiness and credibility of the information source, commonly described as ‘second-hand evaluation’ or evaluating ‘who said it’ (Bromme & Goldman, 2014, p. 65; Sharon & Baram-Tsabari, 2020, p. 879). It can be assessed using the two complementary principles of expertise and reliability especially with the widespread presence of misinformation and unreliable information sources (Allchin, 2023). It is judging the credibility and trustworthiness of an information source by considering who produced the information, where and when it was published, and whether it reflects reliable scientific standards and evidence (Bromme & Goldman, 2014). It also involves evaluating communicators’ expertise through their credentials, publication records, experiences, and alignment with scientific consensus (Duncan et al., 2018; Sperber et al., 2010).

The third aspect, receiver evaluation, emphasises how individuals reflect on the ways information is interpreted. Bielik and Krell (2025, p. 877) described this as ‘third-hand evaluation’, wherein individuals critically examine how prior knowledge, beliefs, experiences, emotions, and sociocultural influences shape judgments of scientific information. Receiver evaluation acknowledges that people are influenced by cognitive biases, political affiliations, and emotional responses when evaluating knowledge claims (Allchin, 2023). Complementing this, the principle of cognitive biases and socioemotional influences recognises that confirmation bias, intuitive reasoning, emotions, social identity, and other sociocultural influences may unconsciously shape the acceptance or rejection of scientific information (Allchin, 2023; Barzilai & Chinn, 2018; Feinstein & Baram-Tsabari, 2024; Tseng et al., 2021). In science education, this aspect encourages students to become aware of factors influencing reasoning and decision-making. In this study, while we acknowledge the broad components that may influence receiver evaluation from literature, we narrow down to students’ personal and anticipated peer understanding and how they information will be accepted. These three aspects and their corresponding principles provide a comprehensive framework for examining how students critically evaluate GenAI-generated scientific explanations (Figure 1).

Epistemic scaffolding for epistemic vigilance

Epistemic scaffolding refers to instructional support that explicitly guides learners’ epistemic thinking and practices by making the underlying epistemology of scientific inquiry visible (Lin & Puntambekar, 2019). Distinct from procedural, conceptual, and metacognitive scaffolding, epistemic scaffolding focuses on how learners evaluate knowledge, justify claims, and engage in disciplinary practices that are aligned with those of scientists. Although conceptually distinct, epistemic scaffolding is rarely implemented in isolation. Instead, it is distributed across instructional materials, classroom discourse, technological tools, and learning activities, where complementary forms of support work synergistically to promote epistemic engagement (Puntambekar, 2021). Within science learning, explicit epistemic scaffolding has been shown to strengthen students’ participation in authentic epistemic practices by making epistemic structures, criteria, and goals accessible to learners (Barzilai et al., 2020; Lin & Puntambekar, 2019).

In this study, epistemic scaffolding is operationalised through the complementary use of epistemic criteria and epistemic question prompts provided by the teacher. The criteria for good scientific explanations, also treated as an epistemic criteria in the study (Figure 1) primarily scaffolded content evaluation by making explicit the disciplinary standards that scientists use to judge the quality of explanations. Instead of relying on intuitive judgments about whether an explanation ‘sounds scientific’, students are provided with explicit standards for evaluating the adequacy of GenAI-generated explanations. As students apply these criteria, they may also begin to question whether the explanation is supported by credible knowledge sources and whether their own understanding influences their interpretation. Thus, while the epistemic criteria are principally directed toward evaluating content, they may simultaneously stimulate source evaluation and receiver evaluation during the evaluation process.

Complementing the epistemic criteria, epistemic question prompts explicitly directed students to examine the credibility of information and to monitor their own epistemic reasoning while interacting with GenAI. Prompts that encourage students to verify explanations with other sources, consider the expertise behind the information, or reflect on whether they understood the explanation explicitly scaffold source evaluation and receiver evaluation. However, these prompts remain grounded in the evaluation of scientific explanations because they are implicitly informed by one or more epistemic criteria. For example, evaluating whether another source confirms an explanation presupposes evaluating the scientific adequacy of the explanation itself, while reflecting on one’s understanding requires consideration of whether the explanation is conceptually coherent and scientifically meaningful. Therefore, the epistemic criteria and epistemic question prompts function as complementary explicit epistemic scaffolds that collectively supported students’ enactment of epistemic vigilance by guiding coordinated evaluation of scientific content, information sources, and their own epistemic reasoning during interactions with GenAI.

Problem statement and research question

This study addresses the gap concerning students’ enactment of epistemic vigilance while co-constructing scientific explanations with GenAI. Specifically, the study examines how two explicit epistemic scaffolds (Barzilai et al., 2020) namely the criteria for good scientific explanations, serving as epistemic criteria (de Andrade et al., 2019), and epistemic question prompts, serving as reflective guides supported students’ enactment of epistemic vigilance (Figure 1). The epistemic criteria directly guided content evaluation while indirectly prompting source and receiver evaluation during interactions with GenAI through specific evaluative criteria. Meanwhile, the teacher-provided epistemic question prompts guided students to reflect on source and receiver evaluation principles while being indirectly informed by at least one epistemic criterion throughout the co-construction process.

In this study, co-construction refers to the complementary roles of students and GenAI in an iterative process of generating, evaluating, revising, and refining scientific explanations. This process included generating images and text, iterative prompting, manual editing of GenAI outputs, and students’ interpretation of these representations to improve their explanations. Within their cognitive levels, students were particularly

focused on iterative prompting, which is one of our classroom norms, known to refine initial results (Oh, 2023; Shin et al., 2024) and the accuracy of GenAI outputs (Chang et al., 2026). Most importantly, students remain the primary epistemic agents, who evaluated the GenAI responses based on the three aspects of epistemic vigilance (Bielik & Krell, 2025; Sperber et al., 2010).

On the other hand, the GenAI complemented the students' epistemic work as an information source and conversational partner that generated scientific explanations, examples, and evidence for students to examine. Most crucial however, was its role as an epistemic provocateur by producing responses that encourage students epistemic vigilance to question, verify, critique. Guided by these framings, the study aims to answer the research question, *'How do middle school science students enact epistemic vigilance when co-constructing and evaluating scientific explanations with GenAI using epistemic scaffolds?'*

Significance of the study

This study advances science education by describing how middle school students enact epistemic vigilance while co-constructing scientific explanations with GenAI through explicit epistemic scaffolds. It extends current discussions beyond GenAI integration to examine students' epistemic vigilance in terms of content, source, and receiver evaluation (Bielik & Krell, 2025; Sperber et al., 2010; Tseng et al., 2021). The study also provides empirical evidence that students' epistemic vigilance may be fostered through explicit scaffolding in the form of an epistemic criteria and epistemic question prompts. It further demonstrates how these scaffolds position students as primary epistemic agents while GenAI functions as an information source, conversational partner, and epistemic provocateur. The findings may also offer practical guidance for educators and curriculum developers seeking to cultivate scientifically grounded reasoning, evaluative judgment, responsible GenAI use, and critical engagement with AI-generated knowledge in contemporary science classrooms through explicit epistemic scaffolding.

Methodology

Context and participants of the study

We used a qualitative research design to describe eight (8) middle school science students' enactment of epistemic vigilance of GenAI-generated knowledge of scientific explanations. The study was conducted in Middle School C in Gyeonggi City, South Korea as a special after-school session rather than a regular graded classroom activity. Prior to participation, it was explained that their participation was not graded, voluntary, and they can withdraw from participation anytime. The students were purposively selected based on three factors. First, they had previously taken the first author's sixth-grade science class at Elementary School S in 2023, which established a strong teacher-student relationship that supported open dialogue and reflective evaluation. Second, they already had experience using GenAI tools, specifically Canva and ChatGPT in earlier science lessons with the first author. This familiarity allowed classroom interactions to focus on epistemic judgment alongside basic tool operation.

Third, they had prior experience constructing scientific explanations using the criteria for good scientific explanations in sixth-grade science. Lastly, they were the only former students attending Middle C which is close to Elementary School S where the first author was currently affiliated during the time of the study, making their participation in the after-school sessions feasible. The purposive sample of four pairs was also appropriate for the close, turn-by-turn analysis required in our qualitative design.

Lesson context

We conducted the study with the two lessons, ‘Why do plants grow well’ and ‘Why is fungus not a plant’ with their agreed classroom norms (Table 1) that positioned GenAI as an information source, a conversational partner, and an epistemic provocateur rather than a substitute for their thinking (Park, 2025).

At the beginning of the lessons, the teacher introduced the criteria for good scientific explanations as explicit disciplinary standards for evaluating scientific explanations as well as provided epistemic question prompts to encourage students to critically evaluate GenAI-generated information by examining its content, source, and how they receive it based on their personal and anticipated peer understanding. Throughout the co-construction process, the teacher consistently reminded the students to use both the epistemic criteria and the epistemic question prompts while interacting with GenAI. Guided by these scaffolds, students were encouraged to critically scrutinise AI-generated explanations, verify information, reflect on their own understanding, formulated increasingly purposeful prompts, and iteratively revise both textual and visual explanations. Through these scaffolding, the students enacted epistemic vigilance by using explicit disciplinary

Table 1. Lessons and classroom norms where students practiced epistemic vigilance while co-constructing scientific explanations with GenAI.

Lesson Topic (Hour)	Target scientific explanation with Gen AI	Classroom norms and description
Lesson 1 (2 h) Why do plants grow well?	Construct a scientific explanation on how plants thrive based on the structure and function of their roots, stems, leaves, flowers, and fruits with Canva	• Joint peer deliberation throughout co-construction <i>Each pair jointly deliberated on every prompt and on each GenAI output; only those ideas that both partners agreed on were carried forward into subsequent prompts.</i>
Lesson 2 (1 h) Why is a fungus not a plant?	Construct a scientific explanation on why mushrooms are not plants with ChatGPT	• Evaluation against the four epistemic criteria <i>Students evaluated each GenAI response against the four criteria of a good scientific explanation.</i> • Iterative reprompting and revision <i>Students treated GenAI outputs as tentative; when an output fell short of the criteria, they reframed their prompts and deleted, revised, or reorganized the outputs accordingly.</i> • Anticipation of peers and reflection on prior knowledge <i>Each pair drew on their own prior knowledge and anticipated how their peers might interpret or counter the co-constructed explanation and adjusted prompts, wording, and representations accordingly.</i>

standards and reflective questioning to evaluate, refine, and justify GenAI-generated scientific explanations.

The teacher asked the students to work in pairs using the provided laptops, which were equipped with Canva and ChatGPT applications, both widely used tools in EdTech-integrated science instruction across Korea (GPOE, 2024). Before the actual conduct of the study, the teacher who is also the primary author facilitated several preparatory lessons focused on the epistemic practice of constructing scientific explanations as a review from their elementary school science classes. These sessions ensured that students were still familiar with the four criteria and could apply them consistently in their epistemic vigilance of their co-constructed scientific explanations with the GenAI.

Data sources

Three forms of qualitative data were collected: video recordings of students' interactions with GenAI, student-generated artifacts, and semi-structured interviews (Figure 2). All classroom conversations and interviews were transcribed verbatim in Korean. While an analysis was conducted on all the original Korean transcripts of the entire dataset, selected excerpts were translated into English for sample data presentation of our findings.

Data analysis procedure

We focused on describing students' practice of epistemic vigilance of GenAI-constructed scientific explanations. The entire dataset, were analyzed following the multi-layered framework (Yin & Zeng, 2025) for qualitative inductive studies (Figure 2). The analytical framework consisted of four interconnected levels: (1) identification and segmentation of epistemic events (macro-level), (2) meso-level coding of evaluative moves, (3) micro-level multimodal discourse analysis, and (4) cross-data interpretation and triangulation. Following this framework, our sample coded data (Appendix 1) show how we merged the transcripts, artifacts, and students' interviews to develop the themes based on the context of our study.

Segmentation and identification of epistemic events (macro-level)

We first applied segmentation to all datasets to capture units of meaningful discourse (Erickson & Shultz, 1992). We identified these segments across the two lessons and compiled them into a running list. We treated each segment to correspond to an episode where students discussed with each other, interacted with, or reflected on their co-constructed scientific explanations with GenAI. We paid particular attention to those segments that captured students' epistemic practices (i.e. prompt constructions and refinement, asking for more detailed illustrations), and evaluation strategies of AI-generated explanations, verify information, reflect on their own understanding. At this level, the video recordings were used to identify episodes in which students explicitly questioned, evaluated, revised, accepted, rejected, or supplemented GenAI outputs while referring to the criteria for good scientific explanations or the teacher-provided

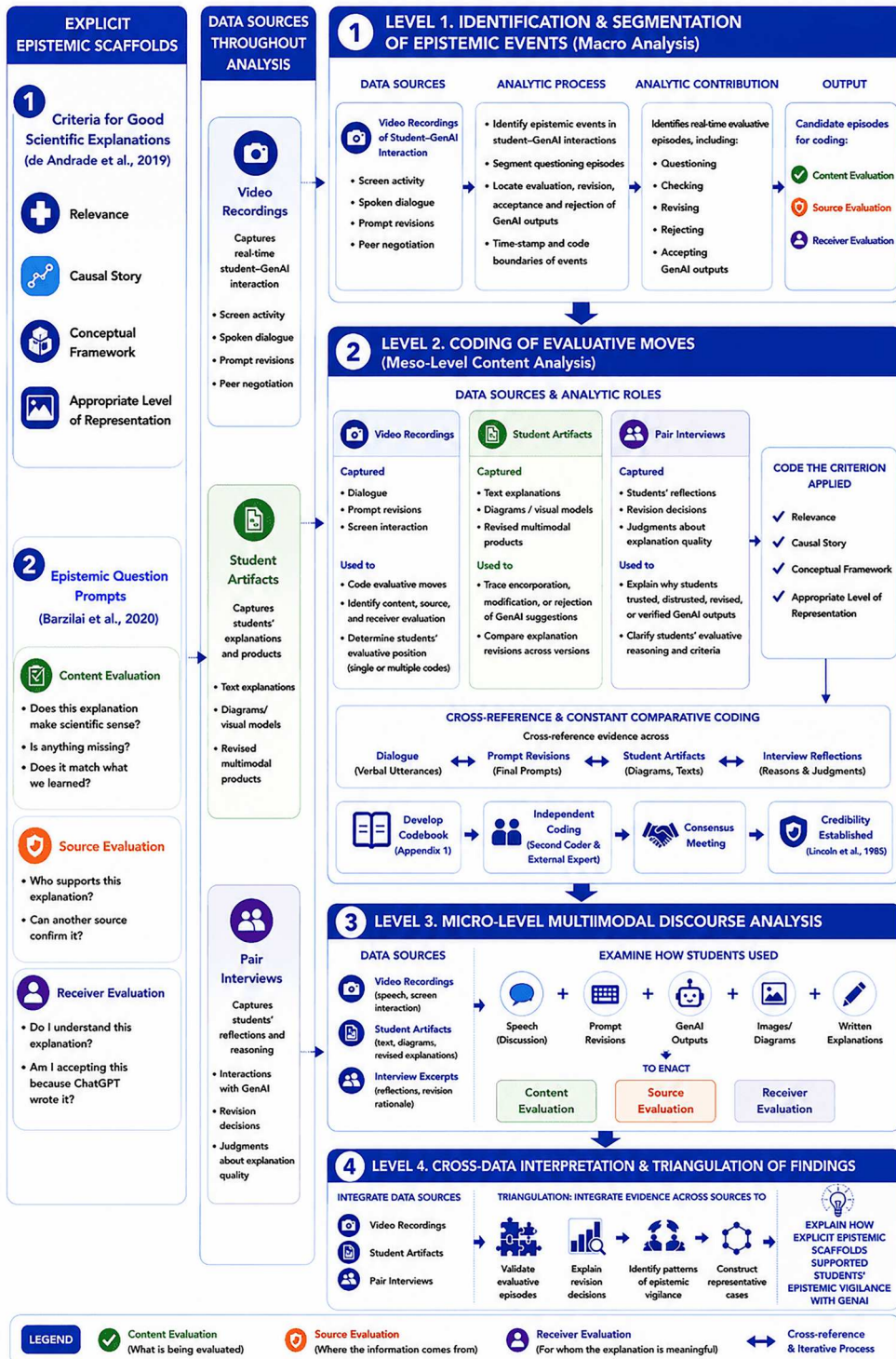


Figure 2. Analytical framework of the study. Note: This diagram design was enhanced by GenAI.

epistemic question prompts. These segmented epistemic events served as the units of analysis for the subsequent coding and discourse analyses.

Meso-level coding of evaluative moves

Using content analysis (Bazerman, 2006), we performed a meso-level analysis on the interaction segments in which students evaluated the GenAI results. We iteratively coded these portions with two objectives. First, we determined the students' evaluative position, determining whether they used content evaluation, source evaluation, receiver evaluation, or all three. Second, we coded how the students used the explicit epistemic scaffolds (Figure 1) when they evaluated the GenAI-constructed scientific explanations. Each segment was categorised using one or more categories, depending on the nature of the students' evaluative assessment. The epistemic criteria served as the primary coding framework for content evaluation, whereas the epistemic question prompts mostly guided the identification of source evaluation and receiver evaluation episodes. Coding was carried out in a constant comparative manner across datasets. Verbal utterances were cross-referenced with their final prompts. Interviews were also compared to their utterances, prompts, and artifacts. Student-generated artefacts were likewise compared across datasets to determine how evaluative decisions were reflected their final co-constructed scientific explanations.

Micro-level multimodal discourse analysis

We conducted a fine-grained, line-by-line discourse analysis (Feng, 2021) on the segmented dataset. This was to unpack how the verbal utterances, written prompts, and artifacts provided meanings on the students' practice of epistemic vigilance as they assessed the accuracy, coherence, and strength of their co-constructed scientific explanations. Using the meso-level codes as analytical anchors, we examined how students enacted content evaluation, source evaluation, and receiver evaluation through their spoken dialogue, prompt revisions, interactions with GenAI, visual representations, and written explanations. The micro-level analysis examined how students enacted content, source, and receiver evaluation during their interactions with GenAI by analyzing the interplay of speech, prompt revisions, GenAI outputs, images and diagrams, and written explanations. Finally, we created a codebook (Appendix 1) detailing how the students practiced epistemic vigilance while being guided by the two explicit epistemic scaffolds. This served as a guide for coding the full dataset, providing us with insights into the principles of epistemic vigilance that the students demonstrated.

Cross-data interpretation and triangulation

We ensured triangulated analysis by iteratively coding all transcripts of students' dialogues, artifacts, and interview transcripts. The three datasets were integrated to explain how the explicit epistemic scaffolds supported students' enactment of epistemic vigilance across the co-construction process. Video recordings captured students' real-time evaluative interactions, student artefacts documented revisions to their scientific explanations, and interviews provided reflective accounts of why particular GenAI outputs were accepted, revised, or rejected. To minimise bias, the second author and an external expert independently coded the data before meeting with the primary author to resolve

discrepancies and reach a consensus; thus establishing the credibility of our study (Lincoln et al., 1985).

Ethicality

Our study was conducted in compliance with ethical guidelines for research, including obtaining informed consent from all participants and their legal guardians and ensuring their confidentiality and anonymity (school is also anonymized). Moreover, we sought the approval of the Gyeonggi Provincial Office of Education, Office of Future Education with Approval No. 981 dated January 18, 2024.

Findings

In the following subsections, we present the themes generated from the analysis. For each theme, we present one representative transcript of the student dialogues containing some artifacts and the corresponding interview responses from each pair of students as illustrative examples of how they enacted epistemic vigilance while co-constructing scientific explanations with GenAI. These examples are intended to illustrate the findings within the context of this study rather than to represent all instances of students' interactions.

Theme 1. Content evaluation of the scientific quality of GenAI explanations through epistemic criteria

Based on the results, the students enacted content evaluation by examining whether GenAI-generated explanations were scientifically plausible, complete, logically coherent, and represented appropriately. Their evaluation was mostly focused on the principles of plausibility, one-sidedness. Rather than accepting GenAI responses as authoritative, they questioned whether explanations were scientifically reasonable, whether important ideas had been omitted, and whether the responses were within the bounds of scientifically accepted ideas. These evaluative practices were primarily guided by the explicit epistemic criteria and supplemented by the epistemic question prompts.

Sample Dialogue 1 illustrates how Students A and B evaluated Canva's explanation of plant growth. After receiving the initial response, they questioned whether it genuinely explained the phenomenon, asking, *'Does this explanation make scientific sense? It gives lots of ideas, but does it explain why plants grow?'* (Line 14). This reflected the criterion of relevance, as students evaluated whether the explanation directly addressed the scientific question. They then shifted their attention to the causal story, with Student A explaining, *'Photosynthesis is important because that's how plants make food. That explains why they can grow'* (Line 15). As they further discussed and recognised that Canva described plant structures separately, Student B suggested revising the prompt to explain *'how they work together'* (Lines 17a–18), applying the criterion of conceptual framework. Their questioning also demonstrated the principle one-sidedness, as they recognised that important causal relationships were missing. So, instead of accepting the initial response, they evaluated it first. Their vigilance was further reinforced by ChatGPT's cautionary statement, *'ChatGPT can make mistakes. Check important info'*,

which served as an epistemic trigger, positioning GenAI not only as an information source but also as an epistemic provocateur that encouraged critical evaluation of its own explanations.

Students also demonstrated plausibility evaluation by questioning whether Canva’s explanation aligned with established scientific knowledge. Student B asked, ‘Does this match what we learned in science class?’ (Line 21), prompting Student A to compare the explanation with their science textbook to determine whether it was scientifically consistent with accepted concepts (Line 22). During the interview, Student B explained that they consulted the textbook because ‘AI can sound correct even when it leaves out important ideas’, indicating that plausibility was judged through its consistency with established scientific knowledge rather than fluency. Student A later reflected, ‘It’s better not to believe it completely just because AI wrote it’ (Line 28), demonstrating awareness that AI-generated explanations should be interpreted within established scientific facts before acceptance. Students also enacted context evaluation by comparing Canva’s explanation with their science textbook (Lines 21–23), and whether the AI-generated explanation reflected accepted scientific knowledge rather than relying solely on its persuasive presentation.

Sample dialogue 1

Line	Student/ GenAI	Dialogue
10	A	So, we have to explain why plants can grow well.
11	B	What question should we ask (to Canva)?
12	A & B	Um... can you just explain why plants can grow so well? I'll put it like this.
13	GenAI	<i>(Canva generates a response explaining that plants grow well because of photosynthesis, roots, stems and leaves, reproductive strategies, and symbiotic relationships with other organisms.)</i>
14	A & B	Photosynthesis, roots, stems, leaves, reproductive strategies, symbiosis... Does this explanation make scientific sense? It gives lots of ideas, but does it explain why plants grow?
15	A	Let's look at them one by one... Photosynthesis is really important... Let's make a video of photosynthesis first... Photosynthesis is important because that's how plants make food. That explains why they can grow.
16	B	Okay. Okay.
17	A	Then there are the roots, stems, and leaves... Can you explain them all at once?
17a	B	The answer talks about each part separately. I think we should ask how they work together.
18	B	That's just explaining the roles... What if we ask it to explain how the interaction between the roots, stems, and leaves helps plants thrive?
19	A	Yes, because if it only lists the parts, people won't understand how the plant survives.

20 GenAI

← Explain how the interactions of the roots, stem, and leaves help a plant grow well.

Plants grow well because their roots, stems, and leaves work closely together to manage the resources needed for life. First, the roots absorb water and minerals from the soil. The stem acts as a passageway, transporting these materials to other parts and providing support. The leaves perform photosynthesis, turning light into sugars the plant needs. Those sugars then move through the stem to the roots and other tissues as energy. Working together in this way helps the plant grow well and adapt to changes in its environment.

AI가 생성한 답변입니다. 텍스트가 적절하지 않은 경우 Canva에 알려주세요.

비슷한 버전

원하는 사항 추가하기



삽입 Enter 키

- 21 B Does this match what we learned in science class?
- 22 A Let's compare it with our science textbook to make sure it's correct.
- 23 B The textbook explains the same process, so Canva's answer seems reliable.
- 24 A Canva didn't tell us where the information came from. That's why we should check another source.
- 25 B This sounds good, but it's a little hard, so let's try it again.
- 26 A Maybe it's hard because we don't understand all the words yet. Let's ask Canva to explain it more simply (*typing...*)
- 27 B Based on other websites, it was actually correct... but it just listed the plant parts... including reproduction strategies and symbiotic relationships...
- 28 A After checking our textbook, I think it's better not to believe it completely just because AI wrote it.
- 29 A Right...we only need simple information on how the plant parts interact with each other for it to grow well...
- 30 B Now our explanation tells what happens and why it happens. That makes it clearer.

Students C and D enacted similar evaluative practices while examining ChatGPT's explanation distinguishing fungi from plants. In Sample Dialogue 2, instead of verifying isolated biological facts, they evaluated whether the explanation sufficiently accounted for fungal classification as a scientific knowledge. Student D recognised that the explanation appropriately connected fungal nutrition with the absence of photosynthesis, demonstrating evaluation of the causal story. However, Student C subsequently observed that the explanation failed to explain that fungi belong to an independent biological kingdom (Line 10). Student D agreed that this information should be added because it made the explanation '*more complete*' (Line 11). These interactions demonstrate students evaluating explanatory completeness through the epistemic criteria of relevance and conceptual framework. They then, recognised that scientific explanations should account for the broader conceptual connections of underlying biological classification rather than presenting isolated characteristics. Here, students again evaluated plausibility by determining whether the explanation accurately reflected accepted biological knowledge. Their observation that ChatGPT omitted fungal classification illustrates attention to one-sidedness, since an important scientific perspective was absent.

Simultaneously, they evaluated the context of the explanation by considering whether it was consistent with established biological classification taught in class, using accepted scientific consensus (Osborne & Pimentel, 2023) which they also reiterated during the interview as a benchmark while judging the plausibility and completeness of the explanation through the criteria for good scientific explanations.

Sample dialogue 2

Line	Student/ GenAI	Dialogue
1	C	We need to summarize why a mushroom is not a plant in our report. How should we ask ChatGPT?
2	D	If we ask the wrong question, it might give too much information. Let's ask for only the main scientific reasons.
3	C	Okay. What should we write?
4	D	[Types] "Tell me the main differences between mushrooms and plants. Give 3–4 scientific reasons why mushrooms are not plants."
5	GenAI	(Summary of ChatGPT's response) Here are four key differences between mushrooms and plants: (1) Mushrooms do not perform photosynthesis; they absorb nutrients from organic matter. (2) Their cell walls contain chitin instead of cellulose. (3) They reproduce by spores instead of seeds. (4) They store energy as glycogen instead of starch.
6	C	Ohhh (Reads the GenAI's answers)...Mushrooms do not photosynthesize because they absorb nutrients from other organisms. That makes sense because plants make their own food, but fungi do not.
7	D	What do we know about the response? Photosynthesis or not, cell wall composition, reproduction method, and energy storage
8	C	Yes, that's what we learned in science class. The explanation matches our textbook.
9	D	It also explains the differences in cell walls, reproduction, and energy storage. Those are real scientific characteristics, not just appearance
10	C	But is anything missing? It doesn't explain that mushrooms belong to a completely different kingdom called Fungi. I think that's an important reason too.
11	D	You're right...we should add that because it makes the explanation more complete...
12	C	Let's also check whether this matches what scientists currently say about fungi.
13	D	Our textbook also says fungi are classified in their own kingdom, so the explanation agrees with current science.
14	C	ChatGPT didn't tell us where it got the information. Let's ask it to provide its sources.
15	D	(Student C typing...)Yes. If it gives information from a biology textbook or a scientific organization, we can trust it more.
16	GenAI	The information is consistent with biology textbooks and scientific references on fungi, including standard biology texts and educational resources from botanical and microbiology organizations.
17	C	Let's also compare it with our science textbook to make sure the information is the same.
18	D	The textbook explains the same four differences, so the information is reliable.
19	C	At first, I thought my prior knowledge that mushrooms were not plants was wrong.
20	D	Me too. I realized I was thinking about what mushrooms look like instead of how they get food and reproduce.
21	C	It was actually correct but... did not mention that it belong to another kingdom.
22	D	Ah, right...good thing we refer to the book... it mentions at the start of its description...

Sample Dialogue 3 illustrates that students applied the same epistemic criteria even when GenAI presented scientific ideas through a story format. After Canva generated a fairy-tale explanation describing how a kidney bean grows, Students E and F questioned whether the story adequately explained the scientific process. Student E asked, *‘Does this explanation make scientific sense? It tells the story, but does it really explain why the plant grows well?’* (Line 30), demonstrating evaluation through the conceptual framework criterion. The students subsequently identified a weakness in the explanation’s causal story. Student F noted that the explanation omitted how water moved from the roots through the stem to the leaves (Line 33), while Student E argued that without this process, readers would not understand how photosynthesis could occur (Line 34).

Guided by the epistemic criteria and epistemic question prompts, they evaluated the explanation based on plausibility and one-sidedness by identifying omitted scientific mechanisms and judging whether the explanation was scientifically coherent and complete rather than merely attractive. During the interview, Student E stated, ‘I remembered our science lesson about stems carrying not only water, so I knew something was missing’, while Student F reflected that ‘just because it’s easy to understand doesn’t mean it’s always complete’ (Lines 42–43). These statements show that students were evaluating the plausibility and completeness of the GenAI-generated explanation.

Sample dialogue 3

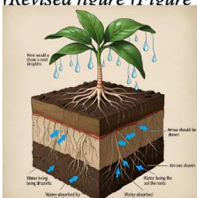
Line	Student/ GenAI	Dialogue
21	E	Let's make a story like a cartoon.
22	F	How? How?
23	E	We make a story as time passes.
24	F	Mmm... So, with the plants we have learned about from science books... How about a kidney bean? A story about how a kidney bean sprouts, grows leaves, stems, and roots, and how they interact with each other to keep it alive.
25	F	Ask Canva this question. <i>"In the form of a story, explain why the kidney bean is able to grow so well. Link cause and effect."</i> (Typing prompts for Canva)
26	GenAI	(Summary of Canva's explanation) Kidney beans first extend roots to absorb water and nutrients; then stems and leaves grow; the leaves photosynthesize to get energy. A fully grown plant then produces flowers, which in turn produce fruit, making seeds.
27	E	I'm going to write this down for Canva. Like this. (Filling in the blanks prompt for Canva) <i>"Roots take up water and nutrients → stems and leaves develop → photosynthesis occurs in leaves → plant grows big enough → flowers and fruits form..."</i> Write a story in this order. Explain why the kidney bean seed grows well and makes a good plant. Explain it as simply as possible in a fairy tale or cartoon format. (Typing prompts for Canva)
28	GenAI	(Summary of Canva's explanation) Here's a simple fairy-tale-style story: (1) In the soft ground, the kidney bean's roots stretched out to drink water and absorb nutrients. (2) Soon, green stems rose, and tiny leaves opened to greet the sun, creating energy through photosynthesis. (3) As time passed, beautiful flowers bloomed, eventually turning into fruits filled with new seeds. (4) The plant grew strong and healthy, ensuring new life in the future.
29	F	OK, OK, we can make a picture or a video clip about it.
30	E	Does this explanation make scientific sense? It tells the story, but does it really explain <i>why</i> the plant grows well?

- 31 F I think it does. The roots absorb water first, then the leaves can make food. That's why the plant keeps growing.
- 32 E Yes, the steps are connected. It isn't just listing the plant parts; it explains how they work together.
- 33 F Is anything important missing? I don't think it explains that the stem carries water and nutrients from the roots to the leaves.
- 34 E You're right. We should add that because otherwise people won't understand how the leaves get the water they need for photosynthesis.
- 35 F Did we learn this before?
- 36 E Yes. The diagram in our textbook also shows the stem transporting water and nutrients.
- 37 F Canva didn't tell us where this information came from.
- 38 GenAI This explanation is based on elementary plant biology concepts commonly presented in science textbooks and educational resources.
- 39 E Can another source confirm this explanation? Let's still compare it with our science textbook because Canva didn't give a specific source.
- 40 F The textbook explains the same process, so the information seems reliable.
- 41 E Do I understand this explanation now? At first, I thought the cartoon story must be correct because it sounded convincing.
- 42 F Me too. But just because it's easy to understand doesn't mean it's always complete.
- 43 E I also remembered our science lesson about stems carrying water, so I knew something was missing.
- 44 F Do we accept this explanation just because Canva made it? I'm glad we checked instead of copying it immediately.

In Sample dialogue 4, students' content evaluation extended beyond textual explanations to multimodal scientific representations. While examining Canva-generated illustrations, Students G and H recognised that the diagram accurately depicted plant structures but inadequately represented the scientific processes occurring within the plant. Student G observed that, '*The roots are touching the soil, but it doesn't really look like water is being absorbed*' (Line 30), while Student H similarly commented that, '*You can't really see water going in*' (Line 31). These observations illustrate evaluation using the appropriate level of representation criterion, judging whether the visual representation effectively communicated scientific ideas. The subsequent request for revisions to include labelled root hairs and directional arrows illustrating water transport, further demonstrated students' commitment to constructing explanations that accurately represented causal mechanisms in visual form. The labelled arrows and root hairs therefore addressed not only representational clarity but also strengthened the completeness of the illustrated explanation.

Guided by the epistemic criteria and epistemic question prompts, these students primarily focused their content evaluation of the illustrated explanation based on one-sidedness before accepting it. Student H questioned whether an important scientific process was missing by asking if the roots should also absorb minerals together with water (Line 41), while Student G explained that omitting this information could lead to an incomplete understanding of plant nutrition (Line 42). They subsequently compared the revised illustration with their science textbook (Lines 43–44) to determine whether it was scientifically consistent with accepted classroom knowledge. During the interview, Student G reflected, 'I remembered our science lesson about water and nutrients moving from the roots to the leaves, so I knew something was missing' (Line 51), indicating that they judged the illustration's plausibility by evaluating whether it accurately and completely represented accepted scientific processes rather than relying on its visual appeal.

Sample dialogue 4

Line	Student/ GenAI	Dialogue
30	G	Look at the picture we made earlier. The roots are touching the soil, but it doesn't really look like water is being absorbed.
31	H	Yeah, it just looks like the roots are sitting in the ground. You can't really see water going in.
32	G	(Typing prompts for Canva) Draw a picture of a root in the soil with the root hairs absorbing water.
33	Canva	(Figure 1 created by Canva) 
34	G	Oh, this is much better. Now it really looks like the plant is taking in water.
35	H	(Typing prompts for Canva) Add arrows to show the direction of water movement from the soil to the root hairs.
36	Canva	(Revised figure (Figure 2) created by Canva) 
37	H	Great. Now let's write next to it: "Water is absorbed into the plant through the root hairs by osmosis."
38	G	Yeah, that makes it clear for anyone looking at it. Just the picture isn't always enough, so the text helps explain it better.
39	H	Does this explanation make scientific sense? Now it explains how the water enters the plant, not just what the roots look like.
40	G	I agree. The arrows show that water moves from the soil into the root hairs and then through the stem to the leaves. That explains why the plant can grow.
41	H	Is anything important missing? Should we also mention that the roots absorb minerals together with the water?
42	G	Good idea. Otherwise people might think the roots only absorb water.
43	H	Does this match what we learned in science class? Let's check our science textbook before we add that.
44	G	(Looking at the textbook) It says that root hairs absorb both water and minerals from the soil, and the stem transports them to the leaves.
45	H	Who supports this explanation? Can Canva tell us where it got this information?
46	Canva	This illustration is based on common plant biology concepts used in educational materials.
47	G	It doesn't tell us the exact source. I trust our science textbook more because it was written by science experts.
48	H	The information in the textbook matches the picture, so I think it's reliable.
49	G	Do I understand this explanation now? At first I thought the first picture was already good enough because it looked nice.
50	H	Me too. But when we looked carefully, I think it didn't really explain what was happening.
51	G	I remembered our science lesson about water and nutrients moving from the roots to the leaves, so I knew something was missing.
52	H	I'm glad we checked instead of choosing the first picture immediately.

From the four dialogues, students appeared to draw upon the same set of explanatory criteria as they consistently evaluated plausibility, one-sidedness, while assessing the GenAI-generated explanations. They also mentioned in our interview that their evaluative moves were further reinforced by GenAI's own warning, 'ChatGPT can make mistakes. Check important info'. Together, these complementary scaffolds encouraged students to suspend immediate acceptance and engage in sustained content evaluation.

Theme 2. Source evaluation by verifying the credibility of GenAI explanations

In this theme, students enacted source evaluation with attention to the reliability and expertise principles (Bromme & Goldman, 2014; Sperber et al., 2010). During their peer discussions and iterative prompting, they sought corroborating evidence, questioned the absence of explicit citations, and compared GenAI outputs with authoritative scientific resources. Within their cognitive levels, their evaluative practices were triggered by the epistemic question prompts while remaining grounded in the epistemic criteria used to judge explanatory quality. Alongside content evaluation, students also appeared to treat ChatGPT's reminder, '*ChatGPT can make mistakes. Check important info*' as an epistemic cue that encouraged them to suspend immediate trust, thus treating this as GenAI's epistemic provocateur function as it constantly prompted them to verify the credibility of its own responses.

For instance, following their evaluation of Canva's explanation of plant growth, Students A and B shifted their attention from the explanation itself to the credibility of the information. Student B asked, '*Does this match what we learned in science class?*' (Line 21) followed by Student A then proposing, '*Let's compare it with our science textbook to make sure it's correct*' (Line 22). After consulting the textbook, Student B concluded that '*The textbook explains the same process, so Canva's answer seems reliable*' (Line 23). They further questioned the transparency of the explanation when Student A observed, '*Canva didn't tell us where the information came from. That's why we should check another source*' (Line 24). These interactions illustrated how they evaluated reliability through corroboration while recognising that trustworthy scientific explanations should be supported by identifiable sources.

Students C and D demonstrated similar source evaluation while examining ChatGPT's explanation distinguishing fungi from plants. After evaluating the scientific completeness of the explanation, Student C suggested, '*Let's also check whether this matches what scientists currently say about fungi*' (Line 12), which was confirmed by Student D in Line 13. Furthermore, they questioned ChatGPT's epistemic authority when Student C remarked, '*ChatGPT didn't tell us where it got the information. Let's ask it to provide its sources*' (Line 14). Student D stated further that '*If it gives information from a biology textbook or a scientific organization, we can trust it more after verifying*' (Line 15), explicitly connecting credibility with disciplinary expertise. Although ChatGPT later stated that its explanation was based on biology textbooks and scientific references (Line 16), students independently verified the information before accepting it (Lines 17–18).

A comparable pattern emerged when Students E and F evaluated Canva's story of plant growth. After determining that the explanation generally aligned with scientific

ideas, Student F asked, *‘Does this match what we learned in science class?’* (Line 35). Student E verified the explanation, noting that it similarly described the transport of water and nutrients (Line 36). They also questioned the transparency of the explanation by asking, *‘Who supports this explanation?’* (Line 37). Although Canva responded that the explanation was based on commonly accepted plant biology concepts (Line 38), Student E remained cautious, suggesting, *‘Let’s still compare it with our science textbook because Canva didn’t give a specific source’* (Line 39). These interactions indicated that in their source evaluation, they mostly paid attention to the scientific verifiable sources that GenAI provided them.

Students likewise extended source evaluation to multimodal scientific representations. After revising Canva’s illustration of water transport in plants, Student H proposed verifying whether the visual explanation agreed with classroom science by asking, *‘Does this match what we learned in science class?’* (Line 43). Students also questioned the origin of the illustration itself. Using the same question prompt used by the other students, Student H also asked, *‘Who supports this explanation? Can Canva tell us where it got this information?’* (Line 45). Although Canva indicated that the illustration was based on common educational materials (Line 46), Student H accepted the illustration only after confirming that it communicated the same scientific process as their reference textbook (Line 48), demonstrating consistent evaluation of reliability and expertise across both textual and visual outputs.

Viewed across the four dialogues, students consistently enacted source evaluation by corroborating GenAI-generated explanations with authoritative classroom resources, questioning missing citations, and considering the expertise underlying scientific claims. Their evaluation was prompted by explicit epistemic question prompts and reinforced by ChatGPT’s own reminder that AI can make mistakes. These complementary scaffolds encouraged students to verify information independently before accepting GenAI-generated explanations, positioning GenAI not only a provider of information but also as a catalyst for critical evaluation of scientific sources.

Theme 3. Receiver evaluation by reflecting on personal understanding

Alongside content and source evaluation, students’ reflection on their own understanding was the primary practice during receiver evaluation. The epistemic question prompts scaffolded them to consider whether they genuinely understood GenAI-generated explanations and whether they accepted them because they supported meaningful understanding rather than simply appearing scientifically convincing.

For Students A and B, receiver evaluation was evident when reflected on whether GenAI-generated explanations made sense to them and that they could later communicate that understanding to others. For instance, after receiving Canva’s initial explanation, Student B questioned whether it matched what they had learned in science class (Line 21), prompting Student A to compare the response with their textbook (Lines 22–24). Student B further remarked that although the explanation ‘sounds good’, it was difficult to understand (Line 25), leading Student A to suggest asking Canva to simplify the explanation because ‘we don’t understand all the words yet’ (Line 26). Student A then emphasised that they needed a simpler explanation of how plant parts interact for plants to grow (Line 29), before Student B concluded that the

revised response was clearer because it explained both what happens and why it happens (Line 30). These interactions demonstrate receiver evaluation with a focus on their personal understanding while being grounded on whether it sufficiently explained the causal process.

Similar practices emerged with Students C and D. During the interview, they explained that the epistemic criteria and question prompts prompted them to compare ChatGPT's explanation with their previous science lessons and textbook. Although they found the explanation scientifically correct, they recognised that it omitted the kingdom classification of fungi, which they considered an important concept. They also noted that GenAI sometimes 'provides complicated ideas but fail to say the basic ones'. Thus, they reflected that if they could not restate the explanation in their own words, they had not fully understood it.

Comparable reflections were observed with Students E and F. While evaluating Canva's explanation presented as a story, Student E questioned whether they truly understood it despite its convincing presentation (Line 41). Guided by the reflective prompts and the appropriate level of representation criterion, they evaluated whether they could explain the same scientific phenomenon independently. During the interview, they emphasised that they should always ensure 'being able to relay a good scientific explanation of the same phenomenon when asked to do so'.

Receiver evaluation likewise extended to multimodal representations. After comparing two GenAI-generated illustrations, Student G observed that the revised image more clearly represented water absorption by the roots (Line 34), while Student H suggested adding arrows to illustrate water movement for better understanding (Line 35). Student H further reflected that the revised illustration explained how water enters the plant rather than merely showing root structures (Lines 37–40). Drawing on prior classroom learning, they compared the illustration with their science textbook and questioned whether it also represented mineral absorption (Lines 41–49). During the interview, both students explained that they deliberately checked whether the illustration represented both observable and microscopic processes because this supported their understanding and this would enable them to explain water transport more clearly to their classmates.

Overall, receiver evaluation was characterised by students' continual reflection on whether GenAI-generated explanations and representations made sense in relation to their prior knowledge and whether their understanding was sufficient to communicate scientific concepts accurately to their peers, with the epistemic scaffolds supporting this reflective process.

Discussion

This study examined how middle school science students enacted epistemic vigilance while co-constructing scientific explanations with GenAI using two explicit epistemic scaffolds: criteria for good scientific explanations and epistemic question prompts. Across the three themes, students' interactions extended beyond retrieving information to evaluating explanatory quality, scrutinising information sources, and reflecting on their own understanding before accepting AI-generated contributions. These findings demonstrate that epistemic vigilance can be deliberately cultivated in GenAI-supported

science learning when instructional scaffolds make disciplinary standards and epistemic practices explicit.

The first contribution concerns the role of explicit epistemic criteria in mediating students' evaluation of scientific explanations. Previous research has shown that learners frequently accept AI-generated responses because they are fluent and always appear plausible (Kasneci et al., 2023; Mollick & Mollick, 2023). In contrast, students in this study evaluated explanations against disciplinary expectations by questioning causal adequacy, identifying conceptual omissions, reorganising to show interrelationships of plant structures, and revising textual and visual representations. These findings suggest that epistemic criteria functioned not merely as assessment tools but as mediational resources that externalised the norms of scientific explanation, enabling students to participate more authentically in disciplinary evaluation. Therefore, epistemic vigilance on their co-constructed explanation with the GenAI through content evaluation became an observable practice that may be explicitly framed as one of the classroom goals in developing students' epistemic agency.

The complementary use of epistemic question prompts extended this evaluation beyond scientific content to include the credibility of information sources and students' own understanding. Students enacted the three dimensions of epistemic vigilance proposed by Sperber et al. (2010) and Bielik and Krell (2025) in an interconnected epistemic processes rather than discrete stages. Also, students frequently moved between evaluating explanatory adequacy, corroborating claims with authoritative sources, and reflecting on how prior knowledge, assumptions, and anticipated peer understanding shaped their evaluations. This dynamic interplay suggests that epistemic vigilance in GenAI-supported learning is best understood as a coordinated process of epistemic regulation supported by explicit instructional scaffolds (Barzilai et al., 2020).

A central theoretical contribution of this study is the positioning of students as the primary epistemic agents within human–AI co-construction. Although discussions of GenAI in education often emphasise its capacity to enhance epistemic understanding (Chang et al., 2026), the present findings shift attention to learners' epistemic work. Students continuously questioned, challenged, verified, revised, and refined GenAI-generated explanations, with the responsibility for determining scientific acceptability remaining with them rather than the technology. This finding reinforces conceptualizations of epistemic vigilance as an active human capacity for evaluating communicated information (Sperber et al., 2010) and extends this perspective by demonstrating how explicit classroom scaffolds enable students to exercise epistemic agency when interacting with GenAI.

The findings also advance current understandings of co-construction between students and GenAI. Based on the results, GenAI functioned as an information source, a conversational partner, and an epistemic provocateur that stimulated further inquiry. Together with the explicit epistemic scaffolds, students in this study became aware of incomplete, overly general, or insufficiently represented responses. This, therefore prompted them to identify explanatory gaps, formulate increasingly precise questions, and iteratively refine both prompts and explanations. Co-construction therefore emerged as a reciprocal epistemic process in which students' evaluations shaped subsequent GenAI responses, while those responses generated new opportunities for disciplinary reasoning. This perspective reframes the educational value of GenAI in the

learning process from producing responses to prompts to provoking productive epistemic engagement.

The findings likewise suggest that prompting should be conceptualised as an epistemic practice rather than a technical skill. Students refined prompts because they first identified conceptual gaps, questioned explanatory adequacy, or recognised representational limitations. Prompt construction was therefore inseparable from scientific reasoning and epistemic evaluation. This implies that prompt engineering in science education should be embedded within disciplinary inquiry Siegle (2025), where increasingly sophisticated prompts emerge from students' application of explicit epistemic criteria instead of technological trial and error.

Finally, this study highlights the importance of explicit epistemic scaffolding for science education in the age of GenAI. Because AI-generated responses are shaped by users' prompts, students require support to formulate purposeful questions, evaluate evidence systematically, and refine explanations iteratively. These scaffolds functioned as mediational resources that supported participation in authentic scientific practices while preserving students' responsibility for epistemic judgment. Therefore, as GenAI becomes increasingly integrated into classrooms, science education should therefore prioritise developing students' epistemic vigilance alongside scientific reasoning and digital literacy. The findings demonstrate that epistemic vigilance is not an automatic consequence of GenAI use but a learnable disciplinary practice that can be intentionally fostered through carefully designed instructional supports.

Conclusion

This study described how middle school students enacted epistemic vigilance during their co-construction of scientific explanations with GenAI through three interconnected practices: evaluating the scientific adequacy of explanations, scrutinising the credibility of information sources, and reflecting on their own understanding before accepting AI-generated knowledge. These practices were supported by explicit epistemic scaffolds that guided students to question, verify, and refine scientific explanations throughout iterative interactions with GenAI. Thus, the findings demonstrate that epistemic vigilance is not an automatic consequence of using AI but a disciplinary practice that can be intentionally fostered through instructional support.

This study also advances current literature by extending epistemic vigilance from a theoretical construct describing how people evaluate communicated information to an instructional framework for GenAI-supported science learning. It further contributes by showing how explicit epistemic scaffolds operationalise content, source, and receiver evaluation during authentic classroom interactions, while positioning students as the primary epistemic agents and GenAI as an information source, conversational partner, and epistemic provocateur. These findings broaden existing discussions of human-GenAI collaboration by emphasising students' epistemic work instead of the capabilities of AI systems.

The findings also suggest that science education should prioritise the development of epistemic vigilance alongside scientific reasoning and digital literacy. Explicit epistemic scaffolds can help students formulate purposeful prompts, critically evaluate AI-generated explanations, and iteratively improve scientific understanding, thereby supporting more meaningful engagement with GenAI.

Limitations

Several limitations should be acknowledged in this study. First, it involved a relatively small sample of participants, which limits the transferability of the findings to broader educational settings. Additionally, participant selection involved students taught by the first author, creating a preexisting teacher-student relationship that may have influenced participants' interactions and interview responses. Further, due to the students' familiarity with the teacher, they have felt inclined to provide socially desirable responses or demonstrate behaviours perceived as academically appropriate. However, to minimise the bias related to power dynamics between the teacher and the students, the classes were non-graded. This is in addition to the several ethical procedures we implemented including voluntary participation, parental consent, and data anonymization. Future studies should therefore involve larger and more diverse participant groups, include researchers independent of classroom assessment processes, and examine epistemic vigilance practices across varied science learning contexts and educational levels. In light of the limitations, the credibility of this study can be attributed to robust qualitative data corpus, which ensured triangulation and a rich understanding of how epistemic vigilance was enacted during co-construction of scientific explanations with GenAI. Further, the fine-grained line-by-line interaction analysis captured students' epistemic vigilance in real time.

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